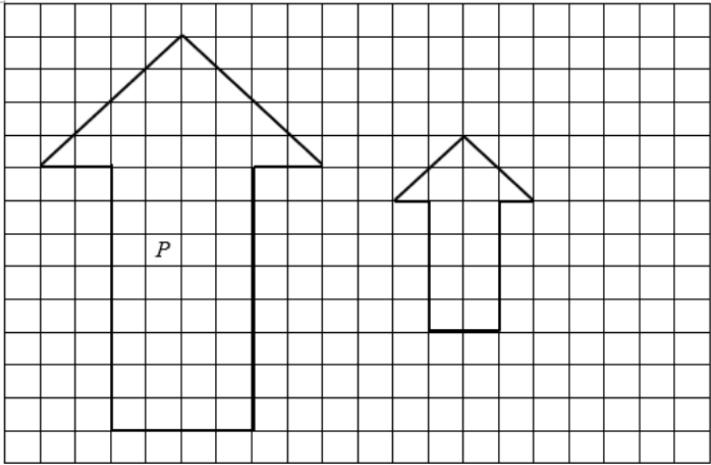


4NA Prelims 2025 Paper 1 (Marking Scheme)

1(a)	$0.0068579 = 0.00686$ (3sf)
1(b)	$125.5\% \times \$320$ $= \frac{125.5}{100} \times 320$ $= \$401.60$
2	$m = 2.058$
	$n = a + b + 1$
3	$3(a + 4b) - 5b - 4a$ $= 3a + 12b - 5b - 4a$ $= -a + 7b$
4(a)	Chocolate and Almond cookies have the same percentage but the size in the pie chart is not the same. OR The orange sector is bigger than the almond sector.
4(b)	This misled readers to believe that chocolate cookies are selling better than almond cookies. OR This misled readers that sales for orange cookies is better than almond cookies sales.
5(a)	$A(-4, 0)$
5(b)	$x = -1$
6	$10.3^2 = 106.09$ $6.4^2 + 8^2 = 104.96$ Since $10.3^2 \neq 6.4^2 + 8^2$, it is not a right-angle triangle. Hence the path will NOT fit nicely into the corner of the garden. OR

	$6.4^2 + 8 = 104.96$ $\sqrt{104.96} = 10.245$ Since $10.245^2 \neq 10.3^2$, it is not a right-angled triangle and the path will NOT fit nicely into the corner of the garden. OR $\cos \theta = \frac{6.4^2 + 8^2 - 10.3^2}{2(6.4)(8)}$ $\theta = 90.632^\circ$ Since $\theta = 90.632^\circ \neq 90^\circ$, it is not a right-angled triangle, and cannot fit nicely around the garden.
7	
8(a)	mean price = $\frac{312}{11} = \$28.36$
8(b)	median price = \$23
8(c)	Median is chosen to represent this data. There is a outlier of \$79 in the data.

9	$6ax - 3ay + 14bx - 7by$ $= 3a(2x - y) + 7b(2x - y)$ $= (2x - y)(3a + 7b)$
10(a)	$9 \text{ km/h} = \frac{(9 \times 1000)m}{(1 \times 3600)s} = 2.5 \text{ m/s}$
10(b)	Distance $= 2.5 \times (45 \times 60)$ $= 6750 \text{ metres}$ OR Distance $= 9 \times \frac{45}{60}$ $= 6.75 \text{ km}$ $= 6750 \text{ m}$
11(a)	$5 - \frac{7}{4}x = 2$ $-\frac{7}{4}x = -3$ $x = 1\frac{5}{7} / \frac{12}{7}$
11(b)	$y = kx^2 \text{ (} k = \text{constant)}$ $3.2 = k\left(\frac{4}{3}\right)^2$ $k = \frac{9}{5}$

	$y = \frac{9}{5}(1.5)^2$ $y = 4\frac{1}{20} / 4.05$
12(a)	$\left(\frac{4b^2}{9a^2}\right)\left(\frac{3a^3}{2b^3}\right)$ $= \frac{2a}{3b}$
12(b)	$5x < 13$ $x < \frac{13}{5}$ $x < 2\frac{3}{5}$ Largest integer = 2
13	$2x = 9 - y \text{ ---- } eq1$ $3x - 2y = 24 \text{ ---- } eq2$ From <i>eq1</i> $2x = 9 - y$ $2x + y = 9$ $4x + 2y = 18 \text{ ---- } eq3 \quad (\times 2 \text{ fr } eq2)$ $3x - 2y = 24 \text{ ---- } eq2$ Eq 3 + Eq 2 $7x = 42$ $x = 6$ $y = -3$

	$\angle BDC = 60^\circ$ (alternate angles $AB \parallel EC$) $\angle BCD = 180 - 60 - 50 = 70^\circ$ ($\angle$ sum of triangle) Since $\angle BDC \neq \angle BCD \neq \angle DBC$, all the angles are not the same. Ryan is not correct . OR $\angle BDE = 120^\circ$ (opp $\angle$ s of // -gram) $\angle BDC = 180 - 120 = 60^\circ$ ($\angle$ s on str. line) $\angle BCD = 180 - 60 - 50 = 70^\circ$ ($\angle$ sum of triangle) Since $\angle BDC \neq \angle BCD \neq \angle DBC$, all the angles are not the same. Ryan is not correct .
18	Let the number of members be x . $\frac{8 + 24 + 18}{8 + 24 + 18 + 15 + 10 + x} = 0.5$ $\frac{50}{75 + x} = \frac{1}{2}$ $100 = 75 + x$ $x = 25$
19	$\text{Height} = 2(2x) = 4x$ Total exterior area $= 2\pi x(4x) + \pi x^2$ $= 8\pi x^2 + \pi x^2$ $= 9\pi x^2$

20(a)	Earnings per cake $= \frac{3600}{120} = \$30$
20(b)	Total delivery time = $18 \times 10 = 180$ mins Total break time (2 breaks, after the 4 th & 8 th delivery) $= 2 \times 15 = 30$ mins Total time taken = $180 + 30 = 210$ mins = $3\frac{1}{2}$ hrs Time complete delivery = 1pm
21(a)	Acceleration $= \frac{35}{20} = 1.75 \text{ m/s}^2$
21(b)	$1.75 = \frac{v - 0}{5 - 0}$ $v = 8.75 \text{ m/s}$
21(b)	Total distance $= \frac{1}{2}(35)(90 + 60) = 2625 \text{ m}$ OR Total distance $= \left(\frac{1}{2} \times 20 \times 35\right) + (60 \times 35) + \left(\frac{1}{2} \times 10 \times 35\right)$ $= 2625 \text{ m}$
22(a)	$1 \text{ cm} : 75000 \text{ cm}$ $1 \text{ cm} : 0.75 \text{ km}$ $4.8 \text{ cm} : 3.6 \text{ km}$ The actual distance is 3.6 km. OR

	$1:75\,000$ $4.8\text{ cm}:360\,000\text{ cm}$ $4.8\text{ cm}:3.6\text{ km}$
22(b)	$1\text{ cm}:0.75\text{ km}$ $1\text{ cm}^2:0.5625\text{ km}^2$ $2.13\text{ cm}^2:1.2\text{ km}^2$ The area on the aviation chart is 2.13 cm^2 . (3sf)
23	$\frac{2}{x-3} - \frac{1}{x+2} = 4$ $\frac{2(x+2) - 1(x-3)}{(x-3)(x+2)} = 4$ $\frac{2x+4-x+3}{(x+2)(x-3)} = 4$ $\frac{x+7}{(x+2)(x-3)} = 4$ $x+7 = 4x^2 - 4x - 24$ $4x^2 - 5x - 31 = 0$ $x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(4)(-31)}}{2(4)}$ $x = 3.48 \text{ or } x = -2.23 \text{ (2dp)}$
24(a)	Area of triangle $= \frac{1}{2}(7.5)^2 \sin 60 = 24.357\text{ cm}^2$ Sector area $= \frac{60}{360} \times \pi (7.5)^2 = 29.452\text{ cm}^2$ Shaded segment Area

	$= 29.452 - 24.357$ $= 5.0954 \text{ cm}^2$ $= 5.10 \text{ cm}^2 (3sf)$
24(b)	$50\pi = 2\pi r^2$ $r^2 = 25$ $r = 5$ By Pythagoras Theorem $13^2 = h^2 + 5^2$ $h = 12$ Total volume $= \frac{1}{3}\pi(5)^2(12) + \frac{2}{3}\pi(5)^3$ $= 575.95$ $= 576 \text{ cm}^3 (3sf)$